

Fundamentals of Artificial Intelligence, Machine and Deep Learning - A.A. 2025/26

v1.3

Deadlines

We remind you that the official course calendar PDF is available in the Materials section on the course portal.

Group Paper Presentation

- Paper presentation dates for each group are available in the following file, in the “Assigned Papers and Presentation Calendar” tab:
[📄 FAIMDL 2025/26 - Groups, Project Topics, and Paper Presentation Calendar](#)
- Each group must provide a **permanent** link to the group presentation slides **no later than the day before (23.59) the scheduled presentation**. Submission instructions will be announced in due time.

Group Project

- **Full project report & code deadline: June 10th (23.59, Italian time)**

Written Exam Calls

Important - it is **strictly mandatory** to individually register for the written exam through the PoliTo student portal, respecting the deadline reported on the portal:

1. **First call:** June 22nd (to be confirmed)
Registration deadline: 16/06/2026 14:00
2. **Second call:** July 8th (to be confirmed)
Registration deadline: 02/07/2026 14:00
3. **Third call:** September 9th (to be confirmed)
Registration deadline: 03/09/2026 14:00
4. **Fourth call:** Winter 2027 (to be confirmed)

Instructions and General Information

- **Reminder:** always check the official communications on the course portal and Discord server.
How to join the server: create a Discord account (with name, surname and PoliTo ID) and following this link: <https://discord.gg/7XXnZaxSEd>
- Course and exam structure slides (already presented in Lecture 1):
<https://file.didattica.polito.it/download/MATDID/34073899?download>

Ethical Code and Plagiarism

All students are expected to adhere to the University's ethical code. If irregularities are found, the exam will be automatically withdrawn, and proper actions will be taken.

- **Plagiarism (report and/or code) will not be tolerated.** Automatic and manual checks will be actively carried out on submitted material
- **Use of GenAI tools for writing the report is not allowed.** You are expected to learn to write a 5-pages report in a group of 4 master's students
- **Any irregularities in the written exam will not be tolerated.** Only pen & paper.

These measures are ultimately aimed at ensuring an effective learning environment.

Exam format and scope

The exam is composed of 3 parts: a group paper presentation, a group project, and an individual written exam.

1) Group Paper Presentation

Groups will be assigned a key research paper in Machine Learning to read and a day to present it in class.

Presentations will take place from April 27th to May 28th in person in class. The full presentations calendar is available in the “Assigned Papers and Presentation Calendar” tab: [FAIMDL 2025/26 - Groups, Project Topics, and Paper Presentation Calendar](#)

The presentation must be compatible with Google slides and must be submitted 1 day before the exam by 23:59. Instructions will be provided for submission.

Follow the instructions on how to read and present a scientific paper (Lecture 2):

<https://file.didattica.polito.it/download/MATDID/34080032?download>

Example of a top-graded paper presentation:

- Presentation: <https://file.didattica.polito.it/download/MATDID/34104963?download>
- Paper: <https://file.didattica.polito.it/download/MATDID/34104969?download>

Important:

- **12 minutes (sharp!) presentation, equally divided between group members (3 minutes per student).**
- **5 minutes for questions**
- No notes are allowed while presenting
- Supplementary materials are not required in the presentation.

Each presenter will receive a score:

- Minimum individual points: 6
- Maximum individual points: 10 (+)
- **Important: Presentation must be sufficient (6/10) for taking the written exam**

Participation is open and warmly suggested to everyone: a curated selection of foundational papers will be presented.

Note: the papers will not be included in the written exam.

2) Group Project

- Minimum group points: 3.5
- Maximum group points: 6
- **Important: Project must be sufficient (3.5/6) for taking the written exam**

There will be 2 project topics:

- Reinforcement Learning
- Image Segmentation

For each topic there will be a series of mandatory objectives (no open project extension). You will work in groups of 4 students.

Project Preference Submission: By Friday, March 13, 2026, you will indicate (one submission per group):

1. Your group composition
2. The project topic you choose

Note for former MLDL students: If you have done (and submitted) the project on topic A for the previous MLDL course, you must be in a group that selects topic B when submitting your group's preference for FAIMDL.

Final report and code submission deadline: June 10th, 2026 - 23:59

1. Report: PDF of **max 5 pages (excluding references)** using the CVPR LaTeX template available [HERE](#)
2. Code submission is mandatory (add link to public GitHub repository in the report, at the end of the abstract)

The reports submitted must motivate and explain all the steps of the assigned project and clearly present and discuss the obtained results. Ask your Teaching Assistants to make sure of the required steps for your project and report format. The report should be self-contained, which means that if some paragraph is referring to something, it should be introduced earlier.

You can keep your submitted project grade until the last exam session (January-February 2027) before the next course begins.

If you decide to use Overleaf for collaboratively working and compiling LaTeX, we suggest creating a shared account for the group, since the maximum number of distinct collaborators in the free version of Overleaf is 2.

Project Report & Code Submission Format:

You have to submit the report using the "Elaborati" section on the course portal. Only one student per group must submit the report as a PDF file on behalf of the group.

- **Important** - Project report PDF file.

Use **exactly** the following format for the file name:

<Group_number>_<RL or IS>_sXXXXXX_sYYYYYY_sZZZZZ_sKKKKKKK
SURNAME1 SURNAME2_ SURNAME3_ SURNAME4.pdf

Example:

99_RL_s123456_s234567_s345678_s456789_Caio_Tizio_Sempronio_Rossi.pdf

- **Project source code:** the code of each project must be uploaded on a public GitHub repository before the final project submission deadline, and a link to the repository must be included in the PDF of the submitted report, at the end of the abstract.

Important: Do not modify the code after the submission deadline.

High-level Project Report Structure

We report below the required structure of the project report in terms of main sections. The Teaching Assistants will provide more specific information for each project on the required contents.

Abstract

- A single-paragraph summary of the report.

1. Introduction

- Project Context: Define the core task
- Literature Foundations: Explicitly cite and discuss the provided references

2. Methodology

3. Experimental Results

- Results reporting: tables and/or plots
- Quantitative Breakdown

4. Discussion

- Result Analysis: Go beyond the numbers and explain why the results happened.
- Methodological Critique: Reflect on the limitations of the approaches

5. Conclusion

- A very short final paragraph summarizing the main outcome.

6. References

- A cleanly formatted list of all literature cited throughout the text, adhering strictly to the report template style.

3) Written Exam

The written exam is individual, and will consist of multiple questions on the basic and advanced topics presented **from “Introduction to Probability” to “3D Learning”**.

- Minimum points: 9
- Maximum points : 15
- **The written exam must be sufficient (9/15) for passing the overall exam**

You will have around 2.5 hours to answer a set of open questions (4 to 7, depending on the topic and complexity).

- Answer in clear and exhaustive text form
- When applicable and helpful, you are also expected to include:

- Pseudocode
- Diagrams
- Main mathematical definitions and statements
- Short examples

In general, the more complete and clear the answers are, the better it is. There is no strict rule, but depending on the case, including formulas, diagrams and/or pseudocode to accompany the text can be very appropriate, useful, and potentially lead to a better grade.

The written exam questions will evenly cover the topics (e.g., key definitions, problem statements, principles, methods, algorithms, architectures, etc.) presented in the lectures listed below:

- L1 Introduction to AI & Machine Learning
- L2: Introduction to Probability
- L3: Learning Theory
- L4: Unsupervised Learning
- L4: kNN, Perceptron, kernels, MLP
- L5: Neural networks optimization
- L5: SVM
- L6: regression, gradient descent, calibration
- L6: optimization recap
- L7 pt1: Introduction to Deep Learning
- L7 pt2: Introduction to Deep Learning
- L8: How to train deep NNs
- L9 pt1: CNN Architectures
- L9 pt2: visualizations
- L10 pt1: Sequences, recurrent neural networks
- L10 pt2: Attention
- L11: Transformers
- Self-Supervised Learning & Generative Models Lecture (April 9th)
- Federated Learning Lecture (April 13)
- Semantic Segmentation Lecture (April 16)
- Reinforcement Learning Lectures (April 20 & April 23)

- 3D Learning Lecture (May 14)

Mathematical proofs are not included, except from the following ones:

- **Theorem:** Bias-Variance Decomposition (*Lecture 2 - Feb26 2026 - IntroProb.pdf*)
- **Intuitive proof:** Hypothesis classes with infinite VC dimension are not PAC-learnable (*Lecture 3 - Mar2 2026 - LearningTheory.pdf*)

Students with officially recognized DSA conditions will have extra time and potentially other measures for the written part and the overall exam. Please make sure to explicitly mention special conditions to the instructor well before the exam starts. Thank you.

Final Grade

- **Written (15) + Project (6) + Paper Presentation (10+)**
- **Maximum grade: 31+**
- **Laude: unanimity of all examiners**

After the paper presentation and project report submission, students will receive a pass/not pass notification for each part. Passing both the paper presentation and the group project is a necessary requirement for taking the written exam.

The final grade will be communicated after the written exam evaluation, taking into account potential bonuses and roundings.

Frequently Asked Questions

Paper presentation

Can I do the paper presentation on a different day than the rest of my group?

No

Can I choose the paper?

No

Can I do the project presentation online?

No

Is the presentation in English?

Yes

What will be asked after the presentation?

Questions on your presentation and the assigned paper may be asked

If my paper presentation is graded as insufficient, can I still take the exam during this academic year?

Yes, in this case each individual student whose presentation has been evaluated as insufficient can ask to be assigned a new paper and will have the opportunity to individually present it (12-minutes presentation) before a chosen future written exam session after the upcoming one.

What are the consequences of failing or rejecting the paper presentation? Do we have to wait a year and do the exam pipeline again?

In this case, upon email request, a new paper would be assigned to the student, with a new date to present individually. Note that if the presentation grade is rejected, also the written exam needs to be done again.

Are we supposed to prepare and present the Supplementary Materials of the paper as well?

No, supplementary materials are not required in the presentation.

Project

Can I propose my own project?

No

How can I do the project alone/in two?

If you are not in a group of 4 students, we'll match you randomly to form a full group

What computing facilities can I use for the project?

Colab

How will I choose the project?

There will be 2 project proposals, each led by a different TA team. There will be ample time for Q&A to help you choose

I am not proficient in Python, will I be able to make the project?

We will point you towards resources and provide 5 basic ML lab sessions in PyTorch, but if you want to make a living in ML you better learn it as soon as possible.

Will I get guidance/help in working on the project?

Yes, during the regular lecture period, TAs will support groups for developing the projects.

Can I reject the grade of a submitted project graded as sufficient?

Yes, you will see the partial grade of the project after the written exam. At that stage, each group member can decide to keep it or reject it. However, if you reject it, you would have to do another project alone from scratch on a different topic, and also redo the written exam. We remind you that project assistance from TAs is only available during the regular lecture period.

If a report is insufficient, will students be still admitted to the written exam?

No. Insufficient projects will have a second chance for revising and resubmitting the project report and code.

Can we modify the code after the report&code submission deadline?

No. The report & code submission deadline implies that the linked repository must not be modified after the deadline, not even readme files.

Is the paper 5 pages maximum excluding the bibliography or including it?

Maximum 5 pages (strict) **excluding** the bibliography.

How should we structure the project report?

In general, the report should provide a comprehensive description of all the aspects of the project, i.e., motivation, related work, problem formulation, methods, experiments, results, etc.

Remember to also add in the report a link to the public GitHub repository containing the project code.

Follow the guidelines provided by your TAs.

Written Exam

How much time will I have?

Around 2.5 hours (plus extra time for officially certified DSA)

Can I have my phone/notes/calculator with me during the exam?

No. You will have to leave any digital or analogue device before sitting the exam. Being found with notes/an electronic device that you forgot to turn in is sufficient evidence to fail the exam and being referred to the disciplinary committee of PoliTo.

I am a student with special needs and the written exam is challenging for me, can I skip it?

No. Signal your need via email well ahead of the exam and we will give you the right setting according to the guidelines of PoliTo.

Are we supposed to know the mathematical formulas related to every one of the exam questions, for example the formula for the Adam optimizer? Or is it enough to describe the concepts with words?

In general, the more complete and clear the answers are, the better it is. There is no strict rule, but depending on the case, including a formula, diagram or pseudocode to accompany the text can be very appropriate, useful, and lead to a better grade.

Mathematical proofs are in general not expected, except the ones specified in the list above.

Can I refuse the grade of the written exam and do it again at the following session?

Yes, you can redo the written exam. The project and paper presentation grades remain valid until the January/February 2027 exam session.

Should we know the size of filters and stride and the number of filters in each convolutional layer of the network like alexnet?

Students are expected to understand and clearly explain the key elements, functioning principles and design choices of the architectures covered in the course. It is expected that the key concepts of stride, receptive field, pooling layer, convolutional layer and filter size are well understood, and simple calculations involving them may be required. Note that the exact filter size is a specifically relevant design choice only for a small subset of the architectures and layers we saw (for example, 1X1 bottleneck layers).

Do we need to include formulas or only text?

There is no strict rule, but depending on the case the motivated inclusion of formulas, diagrams, or pseudocode to accompany the text can be very appropriate, useful, and potentially result in a better grade, since it can demonstrate deeper understanding and clarity of presentation.

Other

Can I reject the final grade and sit again for the exam?

Yes. Note that sufficient project grades and sufficient paper presentation grades remain valid until the last exam session before the following edition of the course starts in the next academic year.

For example, you can decide to redo the written part. In this case, the project and paper presentation report grades remain valid until Jan/Feb 2027.

If I reject the score of the project or the presentation after the written exam on 22nd of June, will I be able to do another project alone (or presentation) for the 2nd written exam call in July? Or only for September?

After the 1st written exam, we will need some days for correcting the test and communicating the final and partial grades. There will not be enough time for individual students to redo the project and/or presentation alone in such a few days. So, if you decide to reject the project or presentation partial grade(s) you would have to prepare it during the summer and then, if passed, do the written exam in the 3rd or 4th exam call.

Instead, if you only reject the written grade, you can redo only the written part, also in the 2nd call.